molecule in equilibrium with the atoms separated by a certain distance. When the light atom is displaced, it will be attracted back to the equilibrium location. The molecule is a miniature version of the weight-and-spring system, but is so small that we have to use quantum mechanics to understand it.

Having worked out the classical Lagrangian, let's try to build a quantum mechanical description of our system. The first thing we need is a space of states. As we've seen, the state of a particle moving on a line is represented by a wave function $\psi(x)$. There are many possible system states, and each one is represented by a different wave function. A function $\psi(x)$ is defined in such a way that $\psi^*(x)\psi(x)$ is the probability density (the probability per unit interval) to find the particle at position x:

$$\psi^*(x)\psi(x) = P(x).$$

In this equation, $P(x)$ represents the probability density. We now have a sort of kinematics—a specification of what the system states are.

Can $\psi(x)$ be any function at all? Aside from the requirement that it must be continuous and differentiable, the only extra condition is that the total probability of finding the particle at any position must be 1:

$$\int_{-\infty}^{+\infty} \psi^*(x)\psi(x)dx = 1. \qquad (10.10)$$

This would not seem to be much of a restriction. Whatever the right-hand side of this equation is, we could always multiply ψ by some constant to make the integral equal to 1—unless the integral is either zero or infinity. Since $\psi^*(x)\psi(x)$ is positive, we don't have to worry about zero, but infinity is a different matter altogether; there are lots of functions that would make the integral in Eq. 10.10 blow up. The conditions for a sensible wave function thus include the requirement that ψ falls to zero fast enough that the integral converges. Functions that meet this condition are called *normalizable*.